

Discussion: The Golden CAPM

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A Remarkable Result

The CAPM is the most influential model in finance — and the most empirically challenged. The dollar CAPM only holds conditionally:

- Tinic & West (1984): only in January
- Cohen et al. (2005): only during low inflation
- Savor & Wilson (2014): only on announcement days
- Hasler & Martineau (2023): in low-volatility states

This paper: denominate returns in ounces of gold, and the CAPM holds unconditionally. Intercept ≈ 0 , slope $\approx$ market return, GRS $p = 0.88$. If correct, this is the first paper to make the unconditional CAPM work.

Replication

Independent replication using WRDS/CRSP confirms the paper's results.

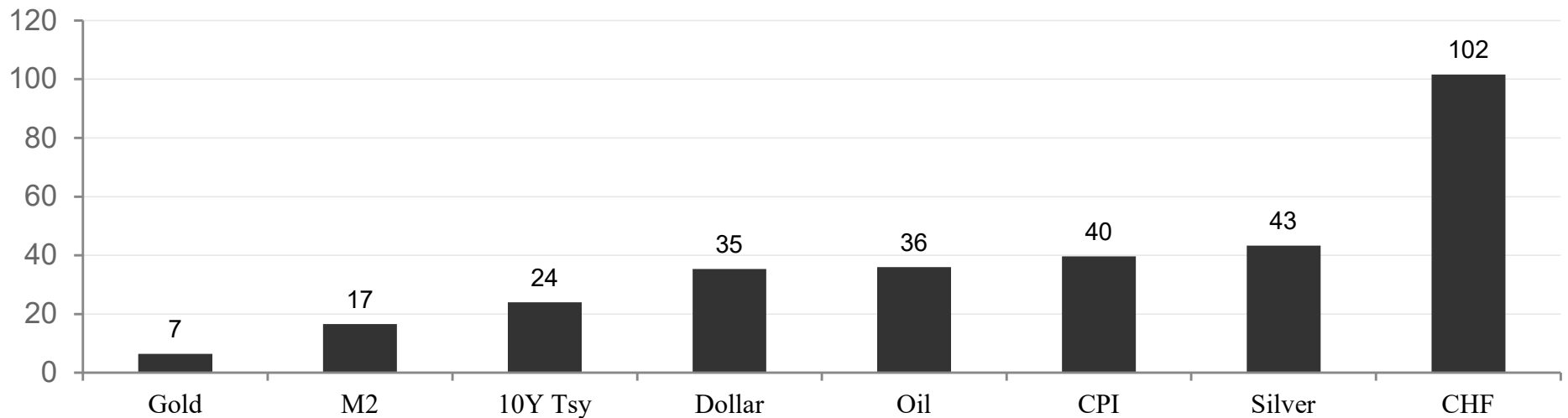
	Intercept	Slope	R ²	FM t(int)	FM t(slope)
Golden (Paper)	3 bps	57 bps	0.72	0.12	1.83*
Golden (Replicated)	3.3 bps	57.1 bps	0.71	0.13	1.83*
Dollar (Paper)	42 bps	28 bps	0.61	3.07***	1.35
Dollar (Replicated)	41.4 bps	27.8 bps	0.62	3.05***	1.36

20 beta-sorted portfolios, July 1927 – December 2024.

Note on pooled regressions: the paper reports pooled $t = 5.20$ for the golden slope vs FM $t = 1.83$. With time-invariant betas, Petersen (RFS, 2009) shows date-clustered pooled SEs equal FM SEs identically. The larger pooled t-stats suggest entity clustering with only 20 clusters. Savor & Wilson (2014), cited as methodological precedent, cluster by date.

Only Gold Works

I tested 7 alternative numeraires and 20+ commodities. None replicates gold's result.



CPI fails (40bp) — not about measured inflation. Swiss Franc worst (102bp) — switching fiat currencies doesn't help. Is gold genuinely special, or does it simply have the right statistical properties?

*Does gold beta capture something economically meaningful —
a priced risk factor that compensates investors —*

*or is it a statistical artifact: a second variable that happens to
have the right properties to fix the SML over this particular
sample?*

The Mechanism

The dollar SML is too flat: slope (28bp) $\approx$ half the market premium (57bp).

Gold returns are uncorrelated with the market ($\rho \approx 0$). Denominating in gold:

1. Leaves the return spread largely unchanged (subtracting gold shifts all portfolios equally)
2. Compresses the beta spread by $1/\omega \approx 2\times$, where $\omega = \text{Var}(R_m)/[\text{Var}(R_m)+\text{Var}(R_{\text{gold}})] \approx 0.47$

Same returns $\div$ half the beta spread = double the slope.

28bp $\times$ 2.1 $\approx$ 59bp $\approx$ golden market premium (58bp) $\rightarrow$ intercept = 0.

This works if and only if:

$$\text{Var}(R_{\text{gold}}) / \text{Var}(R_{\text{mkt}}) \approx (\text{Market Premium} / \text{SML Slope}) - 1$$

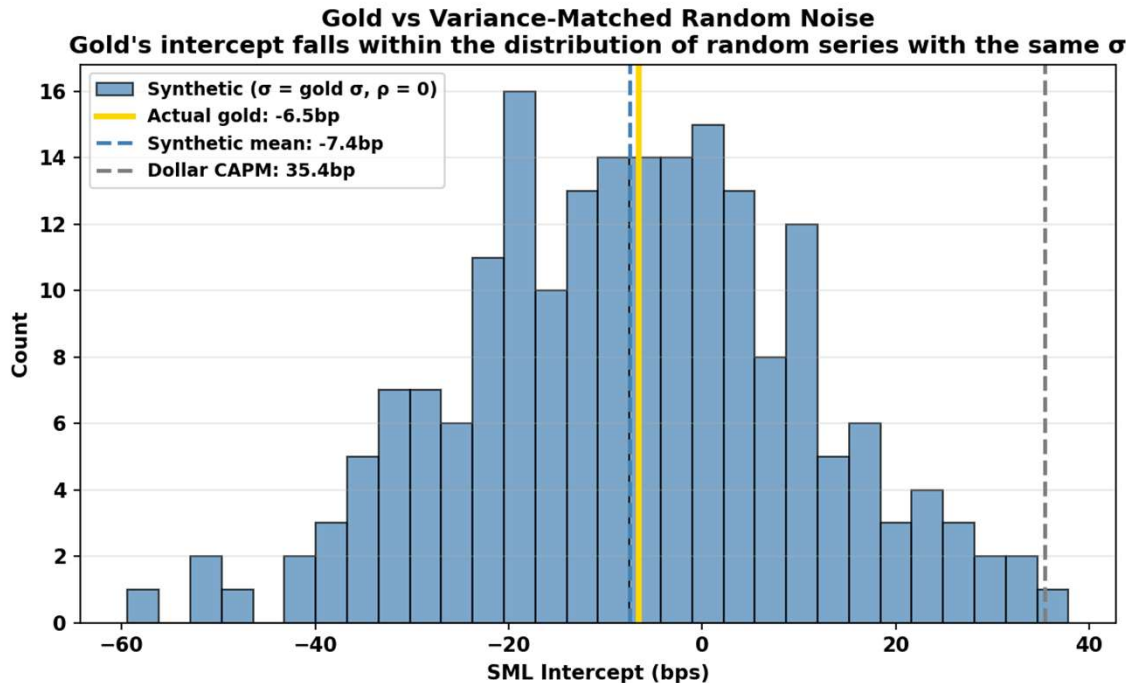
In the data: 1.08 vs 1.01. They match to within 8%.

A Testable Prediction

If the mechanism is purely about the variance ratio, then any random series — pure noise — with gold's mean and variance should produce the same near-zero intercept.

Gold's economic properties (monetary metal, store of value, inflation hedge) might not be causing the results.

Gold vs Random Noise



200 random series, $\sigma = \text{gold's } \sigma, \rho(\text{mkt}) = 0$

Gold (actual): -6.5 bps

Synthetic mean: -7.4 bps
 ± 17.8 bps std

Dollar CAPM: +35.4 bps

Gold's intercept is indistinguishable from random noise with the same variance. Gold's economic properties might not contribute to the result.

Supporting Evidence: Sample Splits

The paper does not test the golden CAPM conditionally. I run 9 splits across 28 subsamples. Gold beats dollar in only 52% of comparisons. 71% of subsamples have $|\text{intercept}| > 30\text{bp}$.

Split	G Int	G Slp/Mkt	\$ Int	\$ Slp/Mkt	Better
Post-1971	5bp	1.03	47bp	0.35	Gold
Up months	-24bp	1.05	35bp	0.88	Gold
Down months	118bp	1.21	33bp	1.12	Dollar
Expansion	-6bp	1.16	44bp	0.53	Gold
Recession	94bp	1.65	13bp	1.06	Dollar
Low inflation	-68bp	1.76	2bp	1.03	Dollar
1980s	477bp	-1.81	178bp	-1.47	Dollar
1990s	9bp	0.97	11bp	1.13	Gold

Golden CAPM intercept ranges from 9bp (1990s) to 477bp (1980s, $t = 2.82^{***}$). The full-sample result averages across decades with large opposing errors.

The authors can reasonably argue: the unconditional CAPM is an unconditional statement. The right test is the full-sample test, and it passes. Subsample variation is expected even under a correctly specified model.

This synthetic test raises a deeper question: is there an economic reason why $\text{Var}(R_{\text{gold}}) \approx \text{Var}(R_{\text{market}})$?

Both assets are priced in dollars — monetary policy and inflation uncertainty add variance to both. Portfolio balance forces could endogenously equalize volatilities across competing stores of value. If the authors can provide a structural model explaining this equality, the Golden CAPM becomes an equilibrium result.

Without such a model, the equality appears sample-specific: the variance ratio ranges from 0.44 to 2.63 across decades.

International evidence would be decisive. Does the golden CAPM hold in UK, Japanese, and European equity markets, where both gold's relative volatility and the BAB anomaly may differ?